Ranga Rohit Nallamolu

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PROFESSIONAL SUMMARY

Data Engineer with **3+ years of experience** building scalable batch and real-time pipelines using **Databricks, Apache Spark, Azure, SAP Datasphere, and SAP Analytics Cloud**. Expertise in **ETL/ELT, Delta Lake, Medallion Architecture, cross-cloud integration, data governance, and Spark optimization**, delivering reliable **financial planning, forecasting, and analytics**.

TECHNICAL SKILLS

Programming Languages: Python, Scala, Java, R

Cloud & Platforms: AWS, Azure, Databricks, Snowflake, SAP Datasphere, SAP Analytics Cloud

Data Engineering: ETL/ELT, Apache Spark, Medallion Architecture, Data Governance, DBT, Airflow, Kafka

AI/ML: LLM Multi-Agent Systems, RAG, LangGraph, Machine Learning

BI & Databases: Power BI, Tableau, PostgreSQL, MySQL

EXPERIENCE

BigBlue InfoTech (Client: McKesson), Dallas, TX **Jan 2025 - Present**

Data Engineer

- Engineered scalable **Databricks PySpark** pipelines using Medallion architecture to ingest and transform finance, transaction, and operational data, **reducing data preparation time by 30%** for actuals, budget, forecast, and reporting datasets.
- Implemented automated data quality and reconciliation checks for nulls, duplicates, schema drift, freshness, and source-to-target balances, **saving 15+ hours per month** in manual validation and troubleshooting.
- Optimized Spark performance on finance pipelines by tuning partitioning, applying caching, and right-sizing Databricks clusters, **reducing job runtime and compute costs by 20%** across P&L, revenue, OPEX, and headcount workloads.
- Integrated Databricks Gold-layer Delta tables with **SAP Datasphere**, building consumption ready views and models that served finance datasets to SAP Analytics Cloud.
- Developed interactive **SAP Analytics Cloud** dashboards for P&L, revenue, OPEX, headcount, margin analysis, and reconciliation, **saving 25+ hours per month** in manual analysis and reporting.

University of Massachusetts Lowell, MA **Dec 2022 - May 2024**

Graduate Data Engineer

- Consolidated Student, Academic, Admissions, and Finance data into **Delta Lake** through scalable **Databricks** PySpark and SQL ETL pipelines, **reducing reporting time by 40%**.
- Resolved duplicated transformations across analytics workloads by implementing a **Bronze/Silver/Gold Medallion Architecture** with **Delta Lake**, improving data consistency and downstream report reliability.
- Resolved slow Spark workloads by applying predicate pushdown, Delta Lake OPTIMIZE, Z-ordering, and small-file compaction, **reducing pipeline execution time by 30%**.
- Mitigated student data security risks by implementing **Databricks Unity Catalog** and RBAC, strengthening governance and supporting **FERPA-compliant** analytics.
- Enabled data-driven decisions on enrollment, retention, and revenue by defining KPIs with Admissions, Finance, IT, and Analytics teams and delivering **Tableau** dashboards.

Graduate Research Assistant

- Built scalable **Azure Data Lake and Azure Databricks** pipelines using **PySpark** and layered **Delta Lake architecture** to process 12 years of shotgun metagenomics data, delivering reliable, analytics-ready datasets for research and machine learning.
- Developed **Random Forest and XGBoost** models in **Python** to predict Type 2 diabetes risk from microbiome features, achieving **85% accuracy** and identifying a strong positive correlation (**r = 0.70**) between *Coprococcus* abundance and diabetes risk.
- Developed interactive **Power BI dashboards** connected to **Azure Databricks** to visualize microbiome patterns and diabetes-related insights, enabling data-driven analysis for healthcare research.

PROJECTS

AI-Driven Stock Trading & Research Platform *website Link*

- Built an AI-powered stock research platform using **Claude Haiku/Sonnet** and multi-agent workflows, with Bull/Bear debate agents and automated risk and stop-loss checks, analyzing 100+ stocks daily and **cutting evaluation time by 90%** and manual monitoring by ~**80%**.

EDUCATION

University of Massachusetts Lowell, MA

Master of Science in Computer Science **Sep 2022 - May 2024**

Christ University Bangalore, India

Bachelor of Technology in Computer Science **Aug 2017 - May 2021**